5G MAG The Media Connectivity Association



OVERVIEW AND UPDATES · BROADCAST AND MULTICAST

Media delivery over 5G Multicast Broadcast Services

Multicast and Broadcast delivery for 5G-NR-based terrestrial and non-terrestrial networks

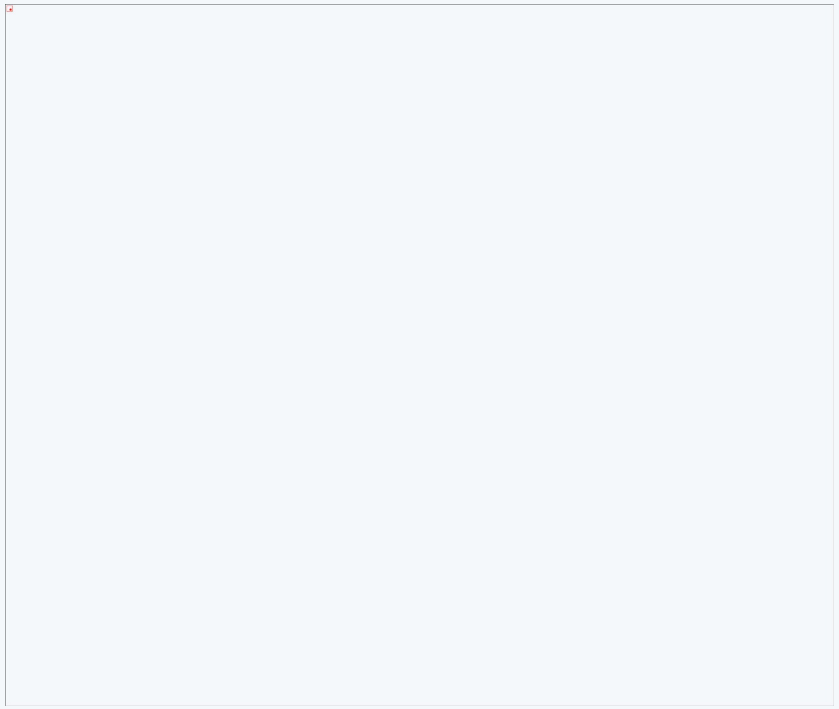
3GPP IETF

NOVEMBER 2026 – www.5g-mag.com

LEARN

About the project

Multicast and Broadcast delivery for 5G-NR-based terrestrial and non-terrestrial networks



MBS User Services network architecture, redrawn from the figure of 3GPP TS 26.502.

3GPP TS 26.502

Technical Specification

The architecture of MBS User Services: the MBSF, the MBSTF and the reference points around them.

3GPP TS 26.517

Technical Specification

The protocols and formats used by MBS User Services.

3GPP TS 23.247

Technical Specification

MBS in the 5G System: the MB-SMF, the MB-UPF and the extensions to existing functions.

3GPP TS 38.300

Technical Specification

MBS in NG-RAN: the NR and NG-RAN overall description, clause 16.10.

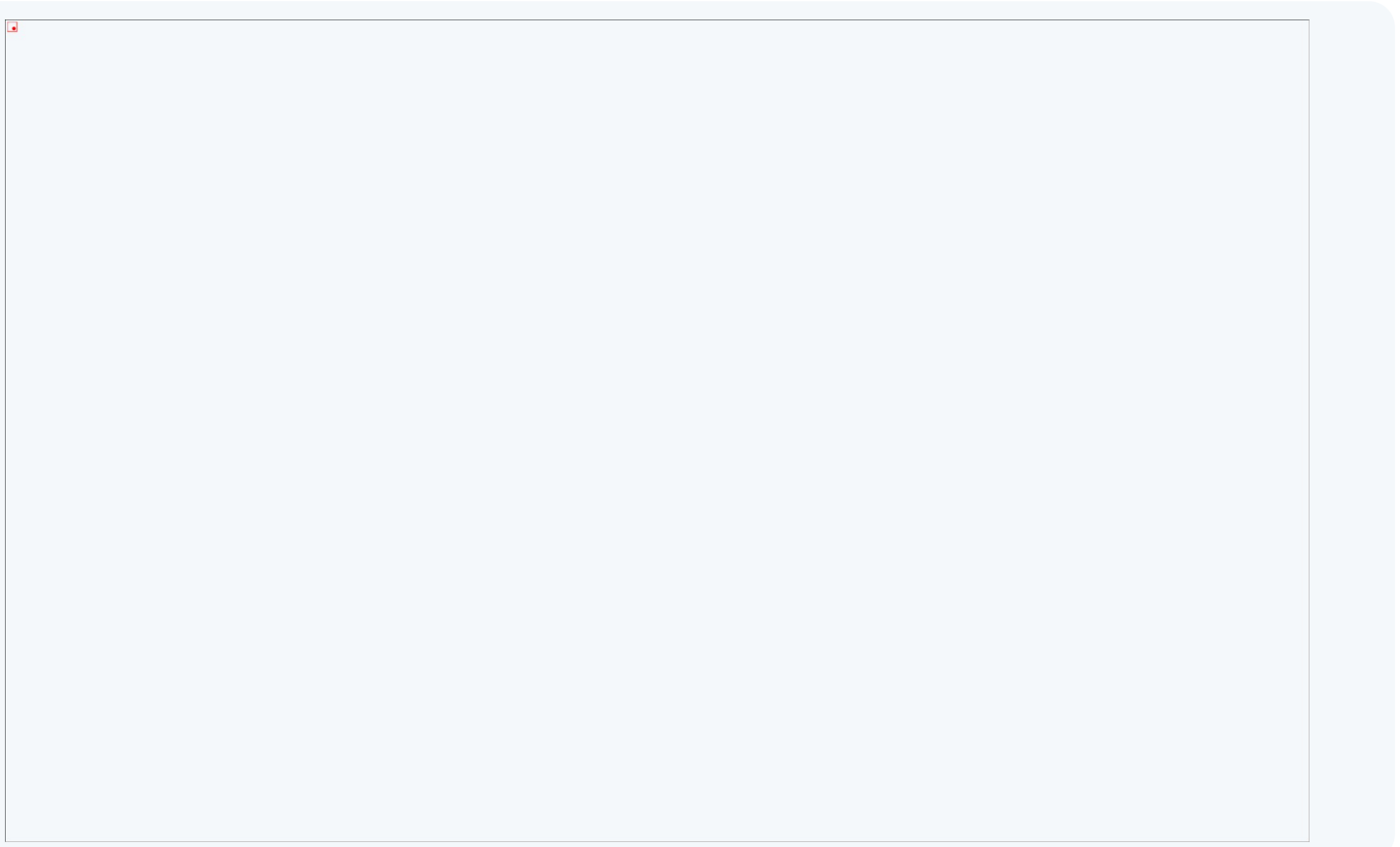
FIND OUT MORE

Technology | Specifications | Reference Tools

LEARN

Broadcast and multicast

Two communication services, and the two ways the 5G Core delivers them



Broadcast

The same content is provided at the same time to all UEs in a geographical area. The content provider and the network may not be aware whether the UEs are receiving it.

Multicast

The same content is provided at the same time to a dedicated set of UEs. The content provider and the network can be aware whether the UEs are receiving it.

5GC Shared delivery

A single copy to each NG-RAN node: used for broadcast and for multicast.

5GC Individual delivery

A separate copy per UE, in a PDU Session: only for multicast.

LEARN

Features: MBS User Services

What the service layer provides

Distribution the MBSTF sends data to the 5G Core with two methods

- Object Distribution Method
- Packet Distribution Method
- Distribution Session control in the MBSTF
- Application Layer FEC
- Time synchronization

Provisioning and announcement the MBSF controls the service and tells the MBS Client about it

- Provisioning of MBS User Services
- User Service Announcement
- Status notifications
- Service areas, location-dependent and multiplexed sessions

Reliability, security and interworking around the delivery

- Object repair
- Security
- Interworking with eMBMS
- Conformance profiles

Reception reporting by the MBS Client and the use of network QoS primitives by MBS User Services are for further study.

LEARN

Features: MBS

MBS in the 5G System and NG-RAN, by the communication service each applies to

MBS Broadcast for all UEs in a geographical area

- Broadcast MBS session
- Frequency selection
- Broadcast for RedCap UEs
- Broadcast over NR NTN
- Resource sharing in network sharing
- Group message delivery

MBS Multicast for a dedicated set of UEs

- Join and leave
- Session activation and deactivation
- Switching shared and individual delivery
- Reception in RRC_INACTIVE

Broadcast and multicast common to both

- Local and location dependent MBS
- QoS and policy control
- Authorisation and identifiers
- Service announcement and pre-configuration
- Power saving
- Interworking with EPS and QoE
- Security
- Roaming (Release 20)

Multicast reception in RRC_INACTIVE was added in Release 18, broadcast over NR NTN in Release 19, access to MBS of a VPLMN while roaming in Release 20.

5G-MAG © 2026

LEARN

How it grew

What each 3GPP release brought to MBS

Release 17

MBS USER SERVICES

TS 26.502 and TS 26.517, after study TR 26.802, with TS 29.580 and TS 29.581

MBS IN THE 5G SYSTEM

Stage 2 (TS 23.247), NR multicast and broadcast, security

Release 18

MBS IN THE 5G SYSTEM

Phase 2 and NR enhancements, RedCap UEs

MULTICAST

Reception in RRC_INACTIVE

UE AND CHARGING

Pre-configuration (TS 24.575), charging (TS 32.279)

Release 19

MBS USER SERVICES

Object repair and time synchronisation

MBS IN THE 5G SYSTEM

Broadcast over NR non-terrestrial networks

REQUIREMENTS

Interworking with digital terrestrial broadcast

Release 20

MBS IN THE 5G SYSTEM

Roaming UEs access MBS of the VPLMN

MBS USER SERVICES

MBSF API enhancements (TS 29.580)

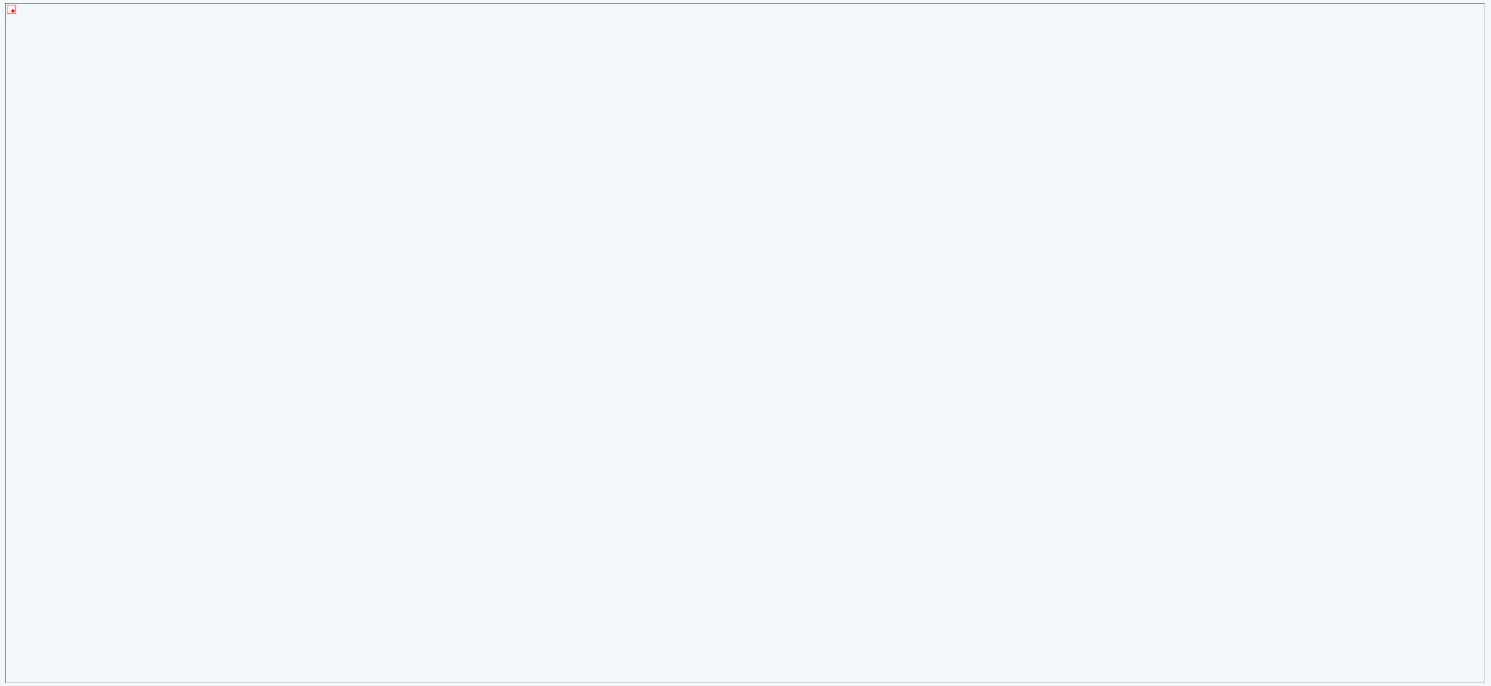
UNVERIFIED

No TR 21.920 description; work plan of 28 Sep 2026

LEARN

Architecture

MBS in the 5G System: functions and reference points



MBSF and MBSTF

The optional service layer. The MBSF is the control plane function; the MBSTF is the user plane function, and sends MBS data packets to the MB-UPF.

MB-SMF and MB-UPF

Added to the 5G Core. The MB-SMF manages MBS sessions and allocates TMGIs; the MB-UPF handles the multicast and broadcast data.

NG-RAN

The decision point for switching between point-to-point and point-to-multipoint delivery.

Without a service layer

MBS transports the data without modifying it.

IMPLEMENT

Repositories

The MBS functions, the 5G Core and the tools

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-mbs-function	the MBS Function (MBSF) of MBS User Services; provides Nmb10	MBSF	1.1.0 3 Jul 2026	GitHub
rt-mbs-transport-function	the MBS Transport Function (MBSTF); provides Nmb2, Nmb8 and Nmb9	MBSTF	1.4.0 3 Jul 2026	GitHub
open5gs	5G-MAG's fork of Open5GS with MBS components, MB-SMF and MB-UPF	MB-SMF, MB-UPF	v0.1.4-5mbs-v2.7.2-o5gs 14 Jul 2026	GitHub
rt-5gc-service-consumers	service consumer libraries; the MBSF uses its MB-SMF client	MB-SMF client library	2.1.1 5 Jun 2026	GitHub
rt-libflute	the FLUTE library that the MBSTF and the MBS Client build on	FLUTE	0.12.4 1 Sep 2026	GitHub
rt-mbs-examples	Docker Compose setup for the MBS network functions, and examples	Deployment	v0.4.0 2 Oct 2026	GitHub

FIND OUT MORE: Technology | Specifications | Reference Tools

● released ○ no release yet

More information at 5g-mag.com/tech/5g-mbs

IMPLEMENT

Repositories: client, RAN and provider

The MBS Client, the application and provider tools, and the srsRAN forks

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE
rt-mbs-client	the UE-side MBS Client: receives announcements and FLUTE objects	MBS Client	On request — Early Access
rt-mbs-application	MBS-Aware Application dashboard: lists services, switches reception on and off	MBS-Aware Application	On request — Early Access
rt-mbs-application-provider	MBS Application Provider web portal: provisions services, pushes content	MBS Application Provider	On request — Early Access
srsRAN_4G_mbs	a fork of srsRAN 4G adapted for 5G Multicast Broadcast Services	UE	v0.1.4-5mbs-srsRAN_4G 14 Jul 2026 — Early Access
srsRAN_Project_mbs	a fork of the srsRAN Project adapted for 5G Multicast Broadcast Services	NG-RAN	v0.1.4-5mbs-srsRAN_Project 14 Jul 2026 — Early Access
rt-media-origin	a media origin server and packager for content delivery testing	Media origin	On request — Early Access

FIND OUT MORE — Technology | Specifications | Reference Tools

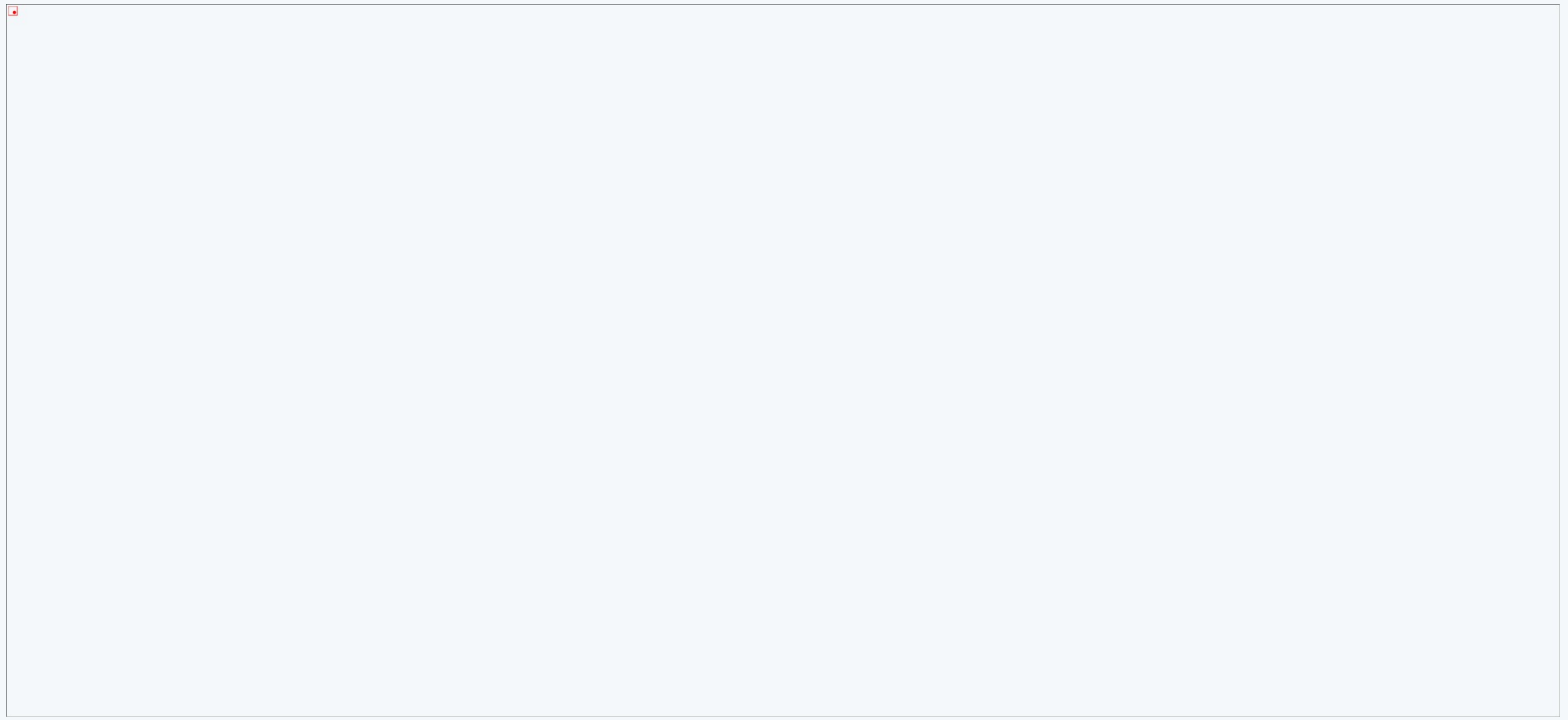
● released ○ no release yet

More information at 5g-mag.com/tech/5g-mbs

IMPLEMENT

Repositories in the architecture

The Reference Tools on the MBS architecture



PROVIDED BY THE REFERENCE TOOLS

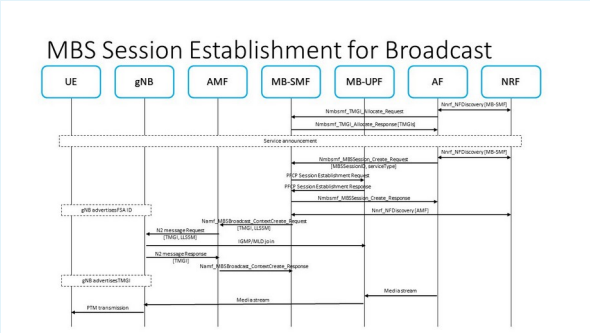
1. **rt-mbs-application-provider**
2. **open5gs**
3. **rt-mbs-function**
4. **srsRAN_4G_mbs**
5. **srsRAN_Project_mbs**
6. **rt-mbs-transport-function**
7. **rt-5gc-service-consumers**
   dependency, used by 3
8. **rt-libflute**
   dependency, used by 6

Boxes without a badge are not mapped to a repository. Badge colour: the project the repository belongs to (orange: 5G MBS; grey: 3GPP RAN and Core Platforms; blue: Content Delivery Protocols and Tools).

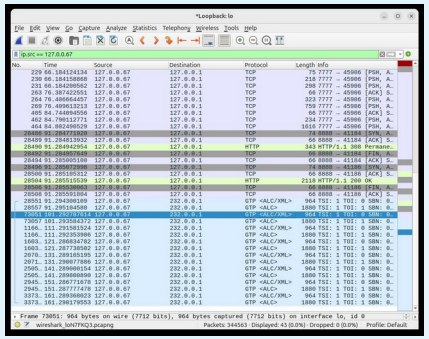
IMPLEMENT

See it and try it

A Broadcast MBS Session on the Reference Tools, and the tutorials that run them



Creating a Broadcast MBS Session



ALC/XML and ALC packets in Wireshark

1 MBS support in the 5GC

Initial implementation of MBS in the Open5GS core, with a Docker deployment.

2 MBS Function and its APIs

The initial implementation of the MBSF: operation and APIs.

3 MBS Transport Function

The initial implementation of the MBSTF, and how to test it.

4 MBS end to end

MBS in Open5GS Core, srsRAN NG-RAN and srsRAN UE, operated together.

WHERE WE STAND

Media delivery over 5G Multicast Broadcast Services

BROUGHT TO STUDY

- MBS Multicast
- MBS over NTN
- Porting to Release 19
- Support in OCUDU

ON-GOING

- MBS Broadcast Rel-18 end-to-end

OUTCOMES

- Implementation of MBS User Services including MBSF and MBSTF
- Initial implementation of 5GC, gNodeB and UE supporting MBS

RELATION WITH SDOS AND INDUSTRY FORA 3GPP IETF

3GPP Feedback logged in the 5G-MAG GitHub project; meeting with the SA4 MBS subgroup on 15 October

5G MAG The Media Connectivity Association

See it. Try it. Contribute.

Raise an issue, send a pull request or join the monthly public call.

GitHub

5g-mag.com

Become a contributor →

CONTRIBUTORS

BBC

Fraunhofer FOKUS

iTEAM Instituto de Telecomunicaciones y Aplicaciones Multimedia